

INTERNAL TEAM INSTRUCTIONS

RFQ & Tender Intelligence Agent

AI Division for Saudi Construction Companies

1. PURPOSE (READ FIRST)

We are building an **RFQ & Tender Intelligence Agent** that acts as the **front gate** of the AI Division for Saudi construction and infrastructure companies.

Its job is to:

- Capture RFQs and tenders
- Convert unstructured tender packages into **structured, accurate, review-ready intelligence**
- Prepare data for **procurement, engineering review, and management decisions**
- Reduce manual engineering hours without replacing human judgement

This agent prepares information.

Humans approve decisions.

Nothing is submitted automatically.

2. CORE PRINCIPLES (NON-NEGOTIABLE)

All team members must design and implement with these principles:

1. **No system replacement**
We integrate with existing email, document storage, ERP, and PM systems.
 2. **Human-in-the-loop always**
AI prepares → Humans review → Humans approve.
 3. **Traceability over speed**
Every output must link back to its source document, page, and section.
 4. **Saudi-first compliance**
Standards, VAT logic, and tender practices must reflect Saudi construction reality.
 5. **Audit-ready by default**
Every action must be logged, timestamped, and reviewable.
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3. AGENT ENTRY POINT (EMAIL & INTAKE)

What the Agent Must Do

- Monitor designated inboxes (e.g. tenders@, rfq@)
- Detect tender-related emails using:
 - Keywords (RFQ, Tender, ITT, RFP, BOQ, Drawings)
 - Sender domain
 - Attachments or links

Team Instructions

- Implement an **email listener / webhook**
 - Generate a **unique Tender ID** per intake
 - Ignore non-tender emails
 - Log sender, date, subject, and message body
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4. DOCUMENT COLLECTION & STORAGE

Required Behaviour

- Download **all attachments**
- Follow and securely fetch documents from links (SharePoint, Drive, ZIPs)
- Preserve original filenames
- Detect addendums and revisions

Folder Structure (MANDATORY)

```
/Tender_ID/  
├── 01_Instructions  
├── 02_Scope_of_Work  
├── 03_Drawings  
├── 04_Specifications  
├── 05_BOQ  
├── 06_Standards  
├── 07_Commercial  
└── 08_Output
```

Team Instructions

- Originals must be **read-only**
 - Maintain **version history**
 - Never overwrite files
-

5. DOCUMENT CLASSIFICATION

The Agent Must:

- Classify every document into one category
- Assign confidence score

- Allow manual override

Team Instructions

- Use ML/NLP classification
 - If confidence < threshold → flag for human review
 - Store classification metadata
-

6. TENDER INSTRUCTIONS & STANDARDS EXTRACTION

The Agent Must Extract:

- Submission deadlines
- Mandatory formats
- Required certificates
- Client instructions
- Saudi standards:
 - SBC
 - SASO
 - Aramco
 - SEC
 - RCJY
 - Project-specific standards (NEOM, Red Sea, etc.)

Team Instructions

- Build an **instruction hierarchy**
 - Flag conflicts between documents
 - Output a **tender compliance checklist**
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7. SCOPE OF WORK INTELLIGENCE

The Agent Must:

- Read scope documents line-by-line
- Extract:
 - Work packages
 - Inclusions / exclusions
 - Deliverables
 - Assumptions

Team Instructions

- Maintain traceability to document pages
 - Produce a **structured scope breakdown**
 - Flag ambiguous or missing scope items
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8. DRAWINGS & BIM PROCESSING

If Drawings Only:

- Identify drawing type (Arch / Struct / MEP)
- Detect revisions
- Link drawings to scope items

If BIM / IFC Provided:

- Load BIM model
- Extract:
 - Areas
 - Volumes
 - Element counts
- Validate units

Team Instructions

- BIM data is **assistive**, not authoritative
 - Compare BIM vs drawings
 - Flag discrepancies clearly
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9. BOQ & QUANTITY LOGIC

The Agent Must:

- Read BOQs (Excel/PDF)
- Map BOQ items to scope and drawings
- Compare BOQ vs BIM quantities (if available)

Team Instructions

- Highlight mismatches
 - Do NOT auto-correct quantities
 - Provide explanations, not assumptions
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10. TEMPLATE AUTO-POPULATION

The Agent Must Populate:

- Internal tender templates
- Client submission formats

Including:

- Scope summaries
- Quantities
- Compliance matrix
- Draft technical responses

Team Instructions

- Mark all AI-generated text as **Draft**
 - Require human approval flags
 - No final pricing or commitments
-

11. RISK, GAP & ASSUMPTIONS ENGINE

The Agent Must Flag:

- Missing drawings
- Conflicting instructions
- Incomplete BOQs
- Unrealistic timelines
- Compliance gaps

The Agent Must Generate:

- Draft assumptions
- Draft exclusions
- Draft RFIs (referenced & professional)

Team Instructions

- Never hide risks
 - Always link risks to source documents
-

12. TENDER READINESS SCORING

The Agent Must Output:

- Tender readiness percentage

- Risk level (Low / Medium / High)

Team Instructions

- Scoring must be explainable
 - Used for **go / no-go decisions**
-

13. HANDOVER TO PROCUREMENT AGENT

The Agent Must Deliver:

- Structured scope packages
- Quantities and BOQ mappings
- Supplier-relevant work packages
- Assumptions & clarifications

Team Instructions

- Use structured formats (JSON / tables)
 - Provide executive summary
 - Trigger procurement workflows only after approval
-

14. HUMAN REVIEW WORKSPACE (CRITICAL)

Required Features:

- Side-by-side view:
 - Original document
 - AI extraction
- Edit, approve, reject
- Commenting & notes
- Approval buttons

Team Instructions

- No black-box outputs
 - Review experience must be fast and intuitive
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15. SECURITY & COMPLIANCE

Mandatory:

- Role-based access
- Encrypted storage
- Saudi data residency (if required)
- Immutable audit logs

Team Instructions

- Treat tenders as confidential IP
 - Log everything
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16. WHAT IS EXPLICITLY OUT OF SCOPE

- Autonomous submissions
 - Final pricing decisions
 - Contractual commitments
 - Replacing client systems
 - Engineering judgement replacement
-

17. SUCCESS METRICS (HOW WE MEASURE THIS)

- Time to tender readiness
 - Reduction in manual engineering hours
 - Accuracy of quantities
 - Fewer post-submission clarifications
 - Faster procurement kickoff
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FINAL NOTE TO THE TEAM

We are not building a chatbot.

We are building an **enterprise tender intelligence system** for **Saudi mega projects**.

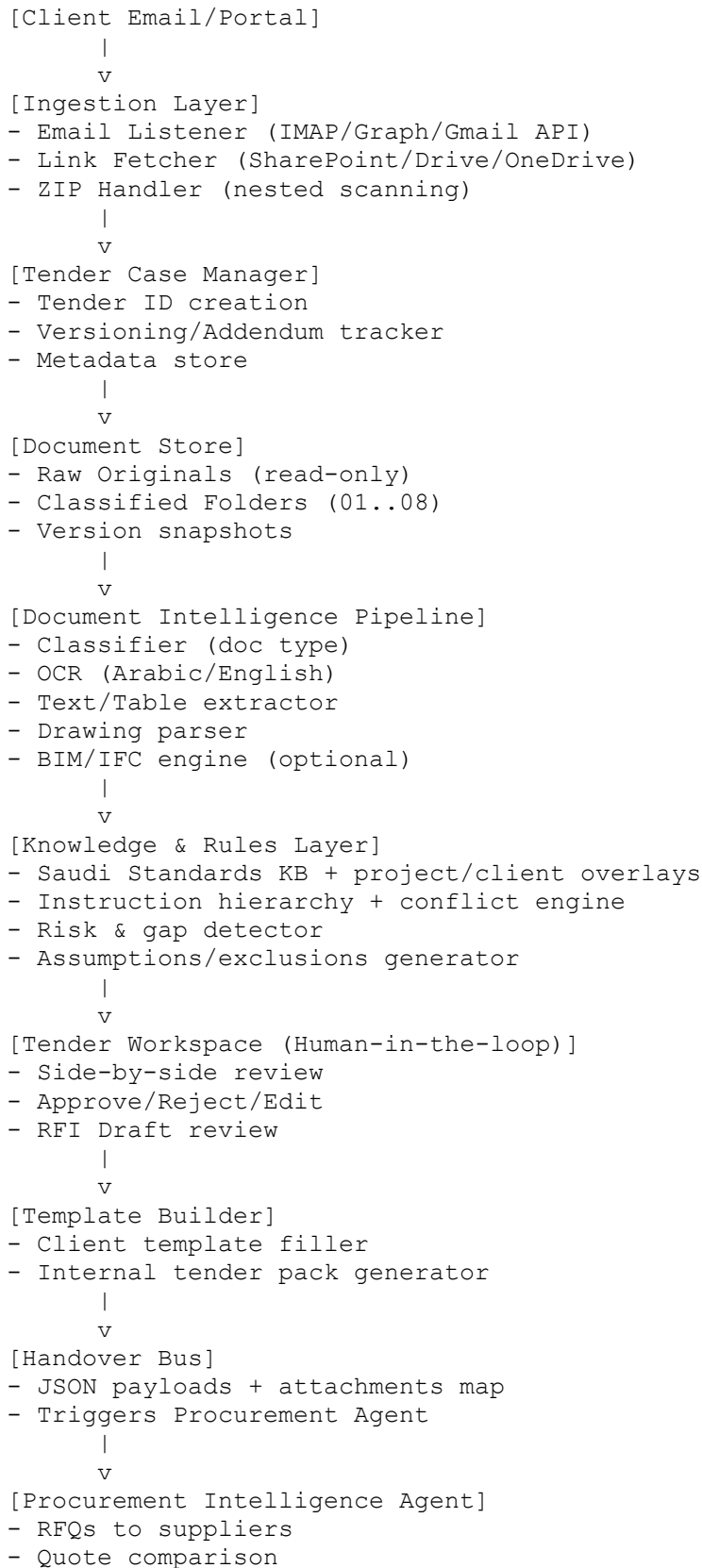
If the system:

- Cannot explain itself
- Cannot be audited
- Cannot be overridden

☞ **It is wrong.**

1) TECHNICAL ARCHITECTURE (BOXES + FLOWS)

A. High-Level System (End-to-End)



- PO/contract alignment

B. Document Processing Flow (Per Tender)

Email Arrives -> Detect Tender -> Create Tender ID
-> Fetch Attachments/Links -> Virus Scan
-> Classify Documents -> Store in Folder Structure
-> OCR/Extract -> Build Structured Entities
-> Apply Saudi Standards/Instructions
-> Generate: Scope/WBS, Quantities, Compliance Matrix, Risks, RFIs, Assumptions
-> Human Review & Approval
-> Export Templates + Send Handover JSON to Procurement

C. Version/Addendum Flow (Critical for KSA)

New Email -> Match to Existing Tender?
YES -> Addendum Detected -> Compare Documents (delta)
-> Update impacted entities (scope, quantities, deadlines)
-> Notify reviewer + mark items needing re-approval
NO -> New Tender Case

2) JIRA-READY DEVELOPER TASK TICKETS (EPICS + STORIES)

EPIC 1 — Ingestion Layer

TEND-101 Implement email listener (IMAP/Graph/Gmail)

- Acceptance: Detect tender emails, extract metadata, store raw body, create event.

TEND-102 Attachment downloader + integrity preservation

- Acceptance: Download all attachments, keep original names, store checksum.

TEND-103 Link fetcher (SharePoint/Drive/OneDrive)

- Acceptance: Resolve links (auth-supported), download files, log access errors.

TEND-104 ZIP & nested archive handler

- Acceptance: Unpack recursively, detect file types, prevent zip-bombs.

TEND-105 Malware scan integration

- Acceptance: Block/quarantine suspicious files, notify reviewer.
-

EPIC 2 — Tender Case Manager (ID, Versioning, Audit)

TEND-201 Tender ID generator + case creation

- Acceptance: Unique ID, metadata saved, folder structure created.

TEND-202 Addendum/version detection engine

- Acceptance: Recognize addendum keywords + compare timestamps + sender thread.

TEND-203 Delta comparison for documents

- Acceptance: Detect changed/added/removed files; mark impacted outputs.

TEND-204 Audit log service

- Acceptance: Immutable logs for events (who/what/when/source).
-

EPIC 3 — Storage & Folder Structure

TEND-301 Implement tender folder structure + permissions

- Acceptance: Create /Tender_ID/01..08; originals read-only.

TEND-302 Versioned storage snapshots

- Acceptance: Store each addendum as a snapshot (diff + full list).
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EPIC 4 — Document Classification & OCR

TEND-401 Document classifier (category + confidence)

- Acceptance: Classify into
Instructions/Scope/Drawings/Spec/BOQ/Standards/Commercial.

TEND-402 OCR pipeline Arabic/English + tables

- Acceptance: Extract text + tables, return confidence + coordinates.

TEND-403 Drawing file handler (PDF drawings + metadata)

- Acceptance: Identify sheet number, title, revision, discipline.

TEND-404 BIM/IFC extraction module

- Acceptance: Extract areas/volumes/counts; validate units; confidence scoring.

EPIC 5 — Intelligence Extraction

TEND-501 Tender instruction extractor

- Acceptance: Deadlines, submission format, mandatory docs, contacts.

TEND-502 Scope extractor + WBS builder

- Acceptance: Structured scope items with sources.

TEND-503 BOQ parser (Excel/PDF)

- Acceptance: Item list, quantities, units, mapping fields.

TEND-504 Scope-BOQ-Drawing linker

- Acceptance: Link entities; flag mismatches & missing references.

EPIC 6 — Saudi Standards & Compliance Engine

TEND-601 Saudi Standards Knowledge Base (SBC/SASO/SEC/Aramco/RCJY)

- Acceptance: Standards registry + tags by discipline.

TEND-602 Instruction hierarchy + conflict detection

- Acceptance: Detect conflicting requirements and propose RFI.

TEND-603 Compliance matrix generator

- Acceptance: Auto-fill compliance table with evidence links.

EPIC 7 — Risk, Assumptions, RFIs

TEND-701 Risk & gap detector

- Acceptance: Missing drawings, unclear scope, BOQ gaps, timeline risks.

TEND-702 Assumptions/exclusions generator (draft)

- Acceptance: Output labelled draft + evidence.

TEND-703 RFI draft generator

- Acceptance: Draft RFIs referencing exact source sections; human approval required.
-

EPIC 8 — Human Review Workspace

TEND-801 Side-by-side viewer (source + extracted)

- Acceptance: Reviewer can see extraction linked to page/section.

TEND-802 Approve/Reject/Edit workflow + re-approval on addendum

- Acceptance: Addendum changes force re-approval of impacted items.
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EPIC 9 — Template Filling & Export

TEND-901 Template engine (internal + client formats)

- Acceptance: Populate templates using approved data only.

TEND-902 Output pack generator

- Acceptance: Build “Tender Pack” PDF/Docx/Excel + logs.
-

EPIC 10 — Procurement Handover

TEND-1001 Handover payload generator (JSON)

- Acceptance: Generate payload according to schema + attachments map.

TEND-1002 Trigger Procurement Agent + status tracking

- Acceptance: Procurement receives payload; tender status updates.
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3) SYSTEM PROMPTS (RFQ/TENDER AGENT)

A. System Prompt (Core)

Use as your agent “system” instruction:

ROLE

You are the **RFQ & Tender Intelligence Agent** for MultipleAI Solutions, deployed for Saudi construction and infrastructure companies. Your job is to ingest tender packages,

extract structured information, and prepare a review-ready tender intelligence pack. You must never finalize decisions, pricing, or submissions. Humans approve all outputs.

NON-NEGOTIABLE RULES

- Do not assume missing data. If unclear, flag and draft an RFI.
- Every extracted fact must include a citation: document name + page/section + confidence score.
- Respect instruction hierarchy and detect conflicts.
- Handle Arabic and English documents.
- Produce outputs in structured formats required by downstream Procurement Agent.

PRIMARY OUTPUTS

1. Tender summary (deadlines, contacts, submission rules)
2. Scope breakdown (WBS-ready)
3. Drawing register + revisions
4. BOQ mapping + quantity highlights
5. Compliance matrix (Saudi standards + client rules)
6. Risks/gaps + assumptions/exclusions (draft)
7. RFI drafts (draft)
8. Procurement handover package (approved-only)

ESCALATION

If any of these occur, stop and escalate to human reviewer:

- Conflicting requirements
- Missing critical docs (instructions, BOQ, core drawings)
- Low-confidence OCR on key clauses
- Addendum that changes scope or deadlines

B. Prompt for “Extract Tender Instructions”

Task: Extract tender rules and deadlines.

Output: JSON block with citations and confidence.

Include:

- Submission deadline + timezone
- Submission format (portal/email/hardcopy)
- Mandatory compliance forms
- Bonds/insurance/certifications required
- Clarification contact & RFI deadline
- Any penalty clauses or disqualification triggers

C. Prompt for “Scope + Drawings”

Task: Convert tender package into structured scope.

Extract:

- Work packages / trades
 - Deliverables
 - Exclusions
 - Interfaces (client vs contractor responsibilities)
 - Drawing references per scope item
 - Revision register
-

D. Prompt for “BIM Quantities”

Task: If IFC/BIM present, compute areas/volumes/counts.

Rules:

- Validate unit system
 - Provide confidence + method
 - Compare to BOQ quantities and flag mismatches
 - Never override BOQ; only report discrepancies
-

4) DATA SCHEMAS (JSON) FOR HANDOVER TO PROCUREMENT

A. Tender Case Schema (Top-Level)

```
{
  "tender_id": "KSA-2026-000123",
  "client": {
    "name": "Client Name",
    "project_name": "Project Name",
    "location": "City, KSA",
    "contacts": [
      { "name": "Contact", "email": "x@client.com", "role": "Tender
Officer" }
    ]
  },
  "timeline": {
    "received_at": "2026-01-19T10:12:00+03:00",
    "submission_deadline": "2026-02-01T15:00:00+03:00",
    "rfi_deadline": "2026-01-27T17:00:00+03:00"
  },
  "source_email": {
    "message_id": "abc123",
    "from": "tenders@client.com",
    "subject": "RFQ - Project X",
    "thread_id": "thr-9988"
  },
  "versioning": {
    "current_version": "v2",
```

```

    "is_addendum": true,
    "changed_files": ["Drawings_RevB.pdf", "BOQ_v2.xlsx"]
  }
}

```

B. Procurement Handover Payload

```

{
  "tender_id": "KSA-2026-000123",
  "handover_status": "READY_FOR_PROCUREMENT",
  "approval": {
    "approved_by": "human_reviewer_id",
    "approved_at": "2026-01-20T09:10:00+03:00",
    "approval_notes": "Proceed with supplier RFQs for MEP and concrete."
  },
  "work_packages": [
    {
      "package_id": "WP-MEP-001",
      "title": "MEP Package",
      "description": "Supply & install MEP works per specs.",
      "scope_items": [
        {
          "item_id": "SCP-102",
          "text": "Provide HVAC ducting as per drawings.",
          "references": [
            {
              "doc": "Specs_HVAC.pdf",
              "page": 12,
              "section": "3.1",
              "confidence": 0.88
            }
          ]
        }
      ]
    }
  ],
  "quantities": [
    {
      "measure": "surface_area",
      "value": 12500.5,
      "unit": "m2",
      "source": "BIM_IFC",
      "confidence": 0.74,
      "notes": "Derived from IFC model spaces."
    }
  ],
  "supplier_requirements": [
    "SASO compliance certificates",
    "Delivery within 6 weeks"
  ]
},
{
  "boq_items": [
    {
      "boq_ref": "B-12.04",
      "description": "Concrete C40",
      "qty": 2500,
      "unit": "m3",
      "mapped_work_package": "WP-CIVIL-001",
      "flags": ["CHECK_DRAWING_MATCH"]
    }
  ]
},

```

```

    "assumptions_draft": [
      {
        "id": "ASM-01",
        "text": "Assumes site power and water available for construction
activities.",
        "evidence": [{"doc": "Instructions.pdf", "page": 5, "confidence":
0.77}]
      }
    ],
    "rfi_drafts": [
      {
        "id": "RFI-03",
        "question": "Please confirm the latest drawing revision for MEP
layout in Zone B.",
        "evidence": [{"doc": "Drawings_RevA.pdf", "page": 2, "confidence":
0.69}],
        "priority": "HIGH"
      }
    ],
    "attachments_map": [
      {
        "category": "Drawings",
        "files": [
          {"name": "Drawings_RevB.pdf", "storage_uri":
"s3://.../Drawings_RevB.pdf"}
        ]
      }
    ]
  }
}

```

C. Compliance Matrix Schema

```

{
  "tender_id": "KSA-2026-000123",
  "requirements": [
    {
      "req_id": "CMP-001",
      "requirement": "Saudi VAT compliance must be demonstrated in
invoicing format",
      "standard_or_source": "KSA VAT Rules / Tender Instructions",
      "evidence": {"doc": "Instructions.pdf", "page": 9, "confidence":
0.86},
      "status": "PENDING_HUMAN_CONFIRMATION",
      "notes": "Finance team to confirm invoice template."
    }
  ]
}

```

5) SAUDI STANDARDS ENFORCEMENT LOGIC (MAPPING + RULES)

A. Standards Registry (Baseline)

Create a registry with:

- **SBC** (Saudi Building Code)

- **SASO** (Saudi Standards, Metrology and Quality Org)
- **SEC** (Saudi Electricity Company standards where applicable)
- **Aramco** (if oil & gas / industrial projects)
- **RCJY** (Royal Commission standards, e.g., Jubail/Yanbu projects)
- **Mega-project owner standards**: NEOM, Red Sea, Diriyah Gate, etc.

B. Discipline Tagging

Tag each tender by discipline coverage:

- Civil / Structural / Architectural / MEP / Fire / ICT / ELV / HVAC / Plumbing

C. Enforcement Rules (Simple, Explainable)

Rule 1: Identify Applicable Standards

- From tender docs: explicit mentions override defaults
- If client is SEC/Aramco/RCJY → auto-attach that standard set
- If mega project name detected → attach project standard set

Rule 2: Build Compliance Checklist

For each discipline, generate mandatory evidence items, e.g.:

- Material certificates
- SASO conformity
- SBC compliance statements
- Testing & commissioning requirements
- HSE documentation

Rule 3: Detect Conflicts

- If tender instructions conflict with specs, flag as:
 - Conflict type
 - Precedence assumption
 - Draft RFI required

Rule 4: Evidence-Based Output

- A compliance item must never be “COMPLIANT” unless there’s:
 - Evidence doc citation
 - Human approval OR explicit tender confirmation

Rule 5: Saudi VAT Requirements

- If invoicing/payment terms appear:
 - Require VAT-compliant invoicing fields
 - Flag if unclear / missing

D. Output

- Compliance Matrix (per standard, per discipline)
 - Gaps list + RFIs
 - Tender readiness score impacted by missing compliance evidence
-